

Kubernetes Setup on Windows (Minikube + Docker) — Notes

Overview

This guide explains how to set up a local Kubernetes environment on Windows using:

- Chocolatey (package manager)
- Docker (container runtime)
- Minikube (local Kubernetes cluster)
- kubectl (CLI tool)

Tools and Purpose

Tool	Purpose
Chocolatey	Install software via command line
Docker	Run containers
Minikube	Create a local Kubernetes cluster
kubectl	Interact with Kubernetes

Installation Steps

1. Install Chocolatey

Run PowerShell as Administrator:

```
Set-ExecutionPolicy Bypass -Scope Process -Force
```

Then install Chocolatey using the official script.

2. Install Required Tools

```
choco install docker-desktop  
choco install minikube  
choco install kubernetes-cli
```

3. Start Docker

- Open Docker Desktop
- Ensure it is running before using Minikube

4. Start Kubernetes Cluster

```
minikube start --driver=docker
```

5. Verify Installation

```
kubectl cluster-info  
kubectl get nodes
```

Expected:

- One node (Minikube)

Deploy First Application

Create Deployment

```
kubectl create deployment myapp --image=nginx
```

Expose Deployment

```
kubectl expose deployment myapp --type=NodePort --port=80
```

Access Application

```
minikube service myapp
```

This opens the application in your browser.

Key Concepts

Cluster

Group of machines running Kubernetes

Node

A machine inside the cluster

Pod

Smallest unit that runs a container

Deployment

Manages pods and scaling

Service

Exposes application to the network

Common Issues

- Docker not running leads to Minikube failure
- Not using administrator PowerShell
- Missing `--driver=docker` flag
- Port conflicts

Summary

- Minikube creates a local cluster
- Docker runs the containers
- kubectl controls the cluster
- You can deploy and expose applications locally

Next Steps

- Learn more kubectl commands
- Understand Pods vs Deployments
- Try scaling applications:

```
kubectl scale deployment myapp --replicas=3
```

Goal

Be able to:

- Run Kubernetes locally
- Deploy applications
- Understand basic architecture